

A Few More Series

1. Find the Maclaurin series for $f(x) = 2x^3 \cos(5x)$.
2. Find the Maclaurin series for $\int 2x^3 \cos(5x) dx$.
3. Find the Maclaurin series for the derivative of $2x^3 \cos(5x)$.
4. Find the first 4 terms of the Taylor series for $f(x) = x \ln x$ centered about $a = 2$.
5. Do the following series converge or diverge? State the test you used. For convergent series with negative terms, indicate if they converge conditionally or absolutely. If appropriate, find the sum.
 - (a) $\sum_{k=1}^{\infty} \left(\frac{k+4}{3k-1} \right)^k$
 - (b) $\sum_{k=2}^{\infty} \frac{k^3 - 3k}{1 + \ln k}$
 - (c) $\sum_{k=3}^{\infty} \frac{1}{k} - \frac{1}{k+1}$
 - (d) $\sum_{k=1}^{\infty} k \frac{3^k}{4^k}$
 - (e) $\sum_{k=0}^{\infty} \frac{(-1)^k}{2\sqrt{k} + 3}$
 - (f) $\sum_{k=2}^{\infty} \frac{3}{k \ln k}$
6. What is the difference between absolute and conditional convergence?
7. Suppose $a_k > 0$ and $\lim_{k \rightarrow \infty} \frac{a_k}{1/k} = \frac{1}{2}$. Does $\sum_{k=0}^{\infty} a_k$ converge or diverge? Why?
8. If $f(x) = \sum_{k=0}^{\infty} \frac{5k}{k!} x^k$, what is $f''(0)$?
9. Indicate if the following are always true or may be false.
 - (a) ----- If $\sum_{k=0}^{\infty} a_k x^k$ converges when $x = 4$, then it must converge when $x = 3$.
 - (b) ----- If $\lim_{k \rightarrow \infty} a_k = 0$, then $\sum_{k=0}^{\infty} a_k$ converges.